

- 1** A programmer is tasked to write a program to store the medical records of a clinic. For each patient, the database would need to store the following data: patient name, identity number, home address, date of visits, diagnosis for each visit, drugs prescribed for each visit, attending doctor for each visit.

The programmer is considering storing this data in either a relational or non-relational database.

- (a)** State two key differences between the two types of databases. [2]
- (b)** State and explain which database system the programmer should choose. [3]
- (c)** State two data obligations the clinic owes its patients. [2]

- 2** Hash tables require the use of hash functions.

- (a)** State two characteristics of a good hash function. [2]
- (b)** Collisions occasionally occur when a hash function generates the same output for two different keys. Describe one method of resolving collisions. [3]
- (c)** The pseudo code on the following page shows an algorithm for an iterative binary search.

Complete the pseudo code by filling in the lines **A**, **B** and **C**. [2]

- (d)** A binary search can be implemented either as a recursive algorithm or an iterative algorithm. Explain how an iterative algorithm might be better than a recursive algorithm in a binary search. [2]

Pseudo code for **2(c)**:

```
01 INPUT SearchValue
02 Found ← FALSE
03 SearchFailed ← FALSE
04 Left ← 0
05 Right ← MaxIndex
06
07 WHILE NOT Found AND NOT SearchFailed
08     Middle ← (Left + Right) DIV 2
09     ....A....
10     THEN
11         Found ← TRUE
12     ELSE
13         IF Left >= Right
14             THEN
15                 SearchFailed ← TRUE
16             ELSE
17                 ....B....
18                 THEN
19                     Right ← Middle - 1
20                 ELSE
21                     ....C....
22             ENDIF
23         ENDIF
24     ENDIF
25 ENDWHILE
26
27 IF Found = TRUE
28     THEN
29         OUTPUT "Value found at location: " Middle
30     ELSE
31         OUTPUT "Value not found"
32 ENDIF
```

3 A bubble sort is a sorting algorithm that works in place.

(a) Name another sorting algorithm that works in place and state its time complexity. [2]

(b) Explain the difference between a sorting algorithm that works in place and one that does not work in place. [2]

(c) The following list of numbers needs to be sorted.

1, 5, 3, 2, 4

Describe the operation of an insertion sort algorithm and show how the listed above can be sorted in ascending order using an insertion sort algorithm. [2]

(d) The following list of numbers needs to be sorted in descending order.

1, 2, 3, 4, 5, 6, 7, 8, 9, 10

A quicksort algorithm needs to select a pivot from the list. It is common to select the first, middle or last item as the pivot. Explain which would be the best pivot in this case. [3]

(e) The following list of numbers needs to be sorted in ascending order.

1, 2, 3, 4, 5, 6, 7, 8, 9, 10

A student claims that insertion sort would require less time to sort the above list when compared to quick sort. Do you agree with the student, and why? [3]

- 4 The following table shows data about a gym's members. The gym has branches at different locations, and members may make a booking for a 1-hour session at any location. Members may also request for a coach during their session.

The booking date/time column has been left blank for reasons of space. A typical entry would be

2022-03-11 09:00.

Member ID	Member Name	Booking ID	Booking Date/Time	Booking Venue	Coach ID	Coach Name
1	Jill Tan	1		Tampines	1	Angela
1	Jill Tan	4		CBD	2	Vicknesh
2	Imran	2		CBD	3	Steven
2	Imran	8		Jurong	3	Steven
3	Sally Khoo	5		CBD	2	Vicknesh
3	Sally Khoo	7		CBD	0	None
3	Sally Khoo	10		Tampines	2	Vicknesh
4	Sam Seng	3		Yishun	1	Angela
4	Sam Seng	6		Yishun	0	None
4	Sam Seng	9		Yishun	1	Angela

- (a) Explain, giving an example, whether the table is in first normal form (1NF). [2]

The gym wants to construct a relational database to store the data shown in the table on Page 5.

The following tables contain the data:

Members	
Member ID	Member Name
1	Jill Tan
2	Imran
3	Sally Khoo
4	Sam Seng

Bookings					
Booking ID	Booking Date/Time	Booking Venue	Member ID	Coach ID	Coach Name
1	2022-03-11 09:00	Tampines	1	1	Angela
2	2022-03-12 12:00	CBD	2	3	Steven
3	2022-03-11 20:00	Yishun	4	1	Angela
4	2022-03-12 12:00	CBD	1	2	Vicknesh
5	2022-03-11 12:00	CBD	3	2	Vicknesh
6	2022-03-13 20:00	Yishun	4	0	None
7	2022-03-12 13:00	CBD	3	0	None
8	2022-03-15 08:00	Jurong	2	3	Steven
9	2022-03-15 20:00	Yishun	4	1	Angela
10	2022-03-13 08:00	Tampines	3	2	Vicknesh

(b) Explain why the table **Bookings** is not in third normal form (3NF). [2]

(c) A table description can be expressed as

TableName (Attribute1, Attribute2, Attribute 3, ...)

The primary key is indicated by underlining one or more attributes with a solid line.

A foreign key is indicated by underlining an attribute with a dotted line.

Write table descriptions for two tables to hold the data from the **Bookings** table, each of which is in third normal form (3NF). [4]

(d) Draw an entity-relationship (ER) diagram showing the necessary three tables and the relationships between them. [4]

(e) Write an SQL query to output the booking dates/times and venues of the member Sally Khoo in chronological order. [5]

(f) Before a coach agrees to a particular booking, the coach needs to check if there is a clash, i.e., the coach already has another booking at the same time, or another booking at a different venue that is too close in time to the requested booking.

Describe how such a clash can be detected. It is not necessary to use SQL syntax, though you may do so. [5]